

**SEMINAR SUMMARY:
APRIL 7, 2026 - RICHARD HAMMACK
GRAPHS, PRODUCTS, AND FOLDING.**

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1. GRAPH PRODUCTS

We began the seminar with the definitions of a graph, as in a graph from graph theory. Let $i, j, n \in \mathbb{Z}^+$, generally a *graph*, G , is a set of *vertices*, $V(G) = \{x_1, \dots, x_n\}$, and a set of *edges*, $E(G) = \{(x_i, x_j) : \text{where } x_i, x_j \in V(G) \text{ and } i, j \leq n\}$. A *bipartite graph* is a graph G with two distinct vertex sets B and W such that no edge joins two vertices of B or two vertices of W , or that edges only traverse from B to W . And, a *direct product of graphs*, G and H is given by

$$V(G \times H) = V(G) \times V(H), \text{ and}$$

$$E(G \times H) = \{(x, u)(y, v) : xy \in E(G), uv \in E(H)\}.$$

An example of a direct product of a graph is drawn in Figure 1. Note this concept of a direct product of a graph was entirely new to me, and was quite fascinating.

We then went about exploring *Weichsel's Theorem* which was given as:

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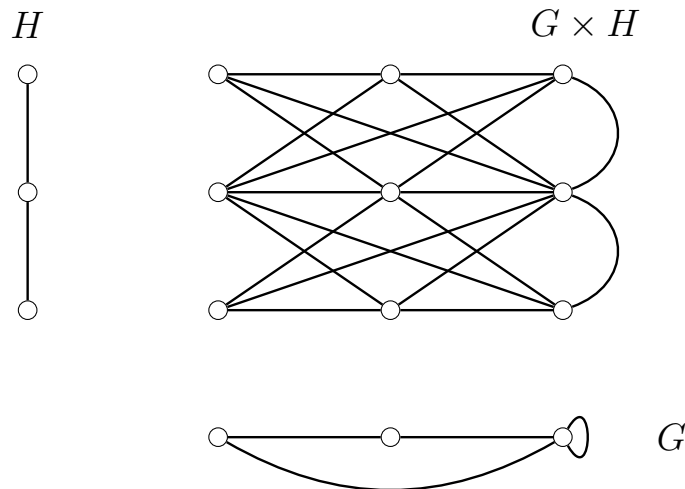


FIGURE 1. A direct product of a path with a cycle G and a path H

Theorem 1. *Suppose that G and H are connected graphs. If both are bipartite, then $G \times H$ has two components. If one has an odd cycles, then $G \times H$ is connected.*

Then we were presented with an interesting conjecture about graph products.

Conjecture 1. *Suppose that G and H are bipartite graphs. Then, $G \times H$ has isomorphic components if and only if G or H has symmetry that reverses its bipartite sets.*

This conjecture was outstanding for quite some time before Hammack was able to make progress and eventually conclude the problem in the affirmative. We were then introduced to a theorem of Ralph McKenzie and Imrich.

Theorem 2. *If G is connected and non-bipartite, then there exists a prime factorization of it, unique up to the order of the factors.*

We subsequently went over how prime factorization of a bipartite graph is not unique, but that there seems to always be a bipartite factor by Weichsel's theorem, however it is unknown as to the uniqueness of that factor. And that is where we left the graph theory to head towards visualizations.

2. POP-UPS AND MULTIDIMENSIONAL VISUALIZATION

After going over the graph direct products, we ventured into the territory of pop-up books and multidimensional visualization. We were shown a series of pop-up books that resented various geometrical figures from a compressed folding. The subtlety here was impressive as we went from a cube to a dodecahedron. When asked, Professor Hammack said that he did the constructions by using tikz and printing a reference graph on card stock and using scissors or a scalpel cut them out then gluing them to the books. All the work had apparently been direct from mind to tikz. We then went in to different visualizations of the four-dimensional cube, the tesseract, some of which I had never thought of before.

I was familiar with the perspective projection of the tesseract, as that is how I personally visualize in four-dimensions. For example think of a cube whose inner colors vary at different cubic depths. Say take the 1m cube, we would color the outside red, and the interior cube with sides of 0.5m if removed would be colored gray. We continue this grading all the way down to the cube of side length ϵ m being black for an arbitrarily small $\epsilon > 0$.

This is similar to how one might think of the 4 dimensional sphere as a sphere with a point source of cold in the middle and an exterior that is hot, and the color grading representing temperature along the interior of the sphere. The projections of the tesseract that I found most interesting were the linear projection and the orthogonal projection through opposite edges. We finished the talk with a pop-up representation of the 5 dimensional cube as a linear projection into 3-space.